A neural circuit theory with recurrent normalization explains the dynamics of visual working memory

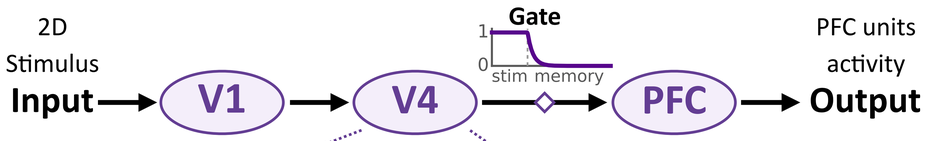
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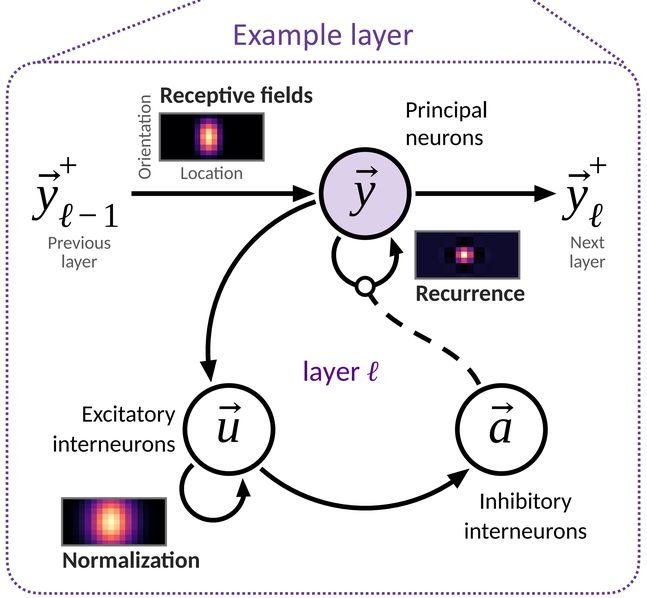
QR code

Motivation: Visual working memory (WM) precision declines with delay, set size, and shorter stimulus duration, but no biologically grounded circuit has explained all of them together.

Conclusion: We developed a neural circuit theory that implements divisive normalization through its recurrence to fit psychophysical measurements of visual working memory dynamics.

ORGaNICs: stochastic neural circuit model[5, 6]





Membrane potential

$$\tau_y \dot{\mathbf{y}} = -\mathbf{y} + b_y \mathbf{z} + \frac{1}{1+\mathbf{a}^+} \odot \mathbf{W}\sqrt{\mathbf{y}^+} + \boldsymbol{\eta}_y$$

$$\tau_u \dot{\mathbf{u}} = -\mathbf{u} + (b_u \sigma)^2 + \mathbf{N}\left[\mathbf{y}^+ \odot (\mathbf{u}^+)^2\right] + \boldsymbol{\eta}_u$$

$$\tau_a \dot{\mathbf{a}} = -\mathbf{a} + \mathbf{u}^+ + \mathbf{a}^+ \odot \mathbf{u}^+ + \alpha \dot{\mathbf{u}} + \boldsymbol{\eta}_a$$

Firing rate

$$\tau_y \dot{\mathbf{y}}^+ = -\mathbf{y}^+ + \lfloor \mathbf{y} \rfloor^2$$

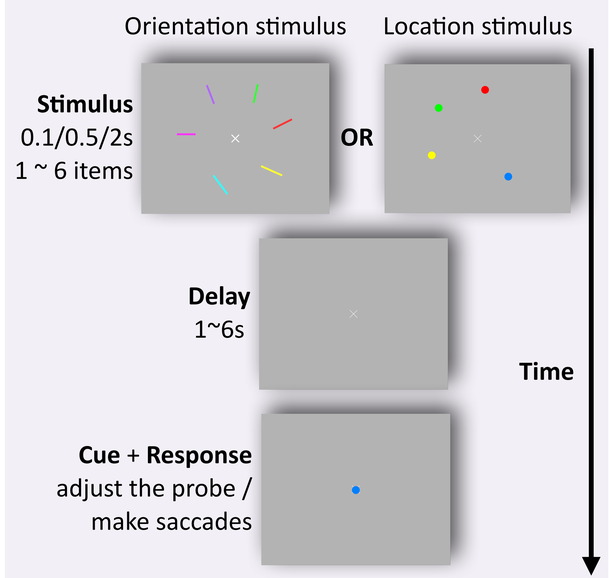
$$\tau_u \dot{\mathbf{u}}^+ = -\mathbf{u}^+ + \sqrt{\lfloor \mathbf{u} \rfloor}$$

$$\tau_a \dot{\mathbf{a}}^+ = -\mathbf{a}^+ + \lfloor \mathbf{a} \rfloor$$

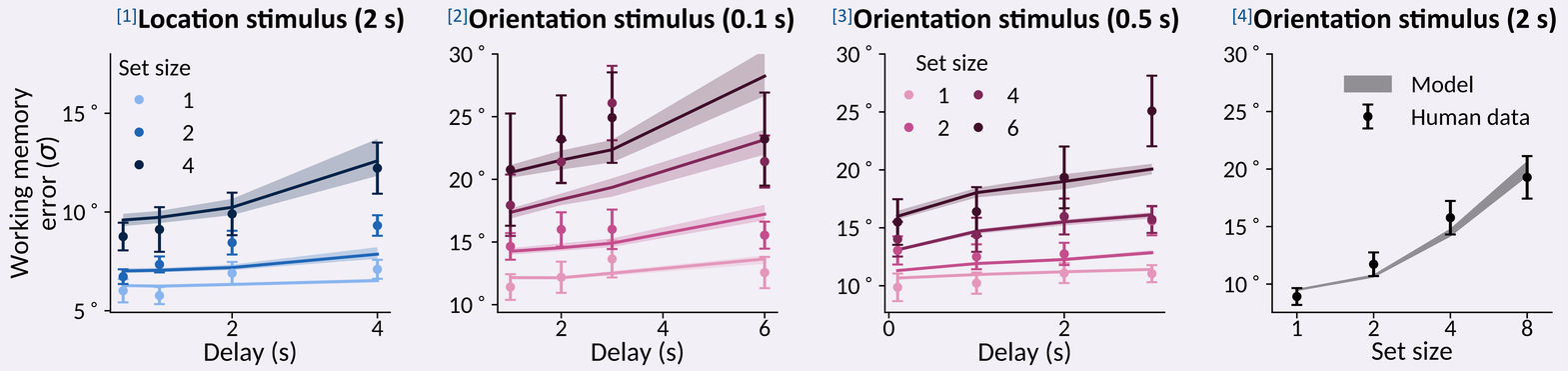
Low-pass filtered noise

$$\tau_\eta \dot{\boldsymbol{\eta}} = -\boldsymbol{\eta} + \rho\sqrt{\tau_\eta}\ \mathcal{N}(0, dt)$$

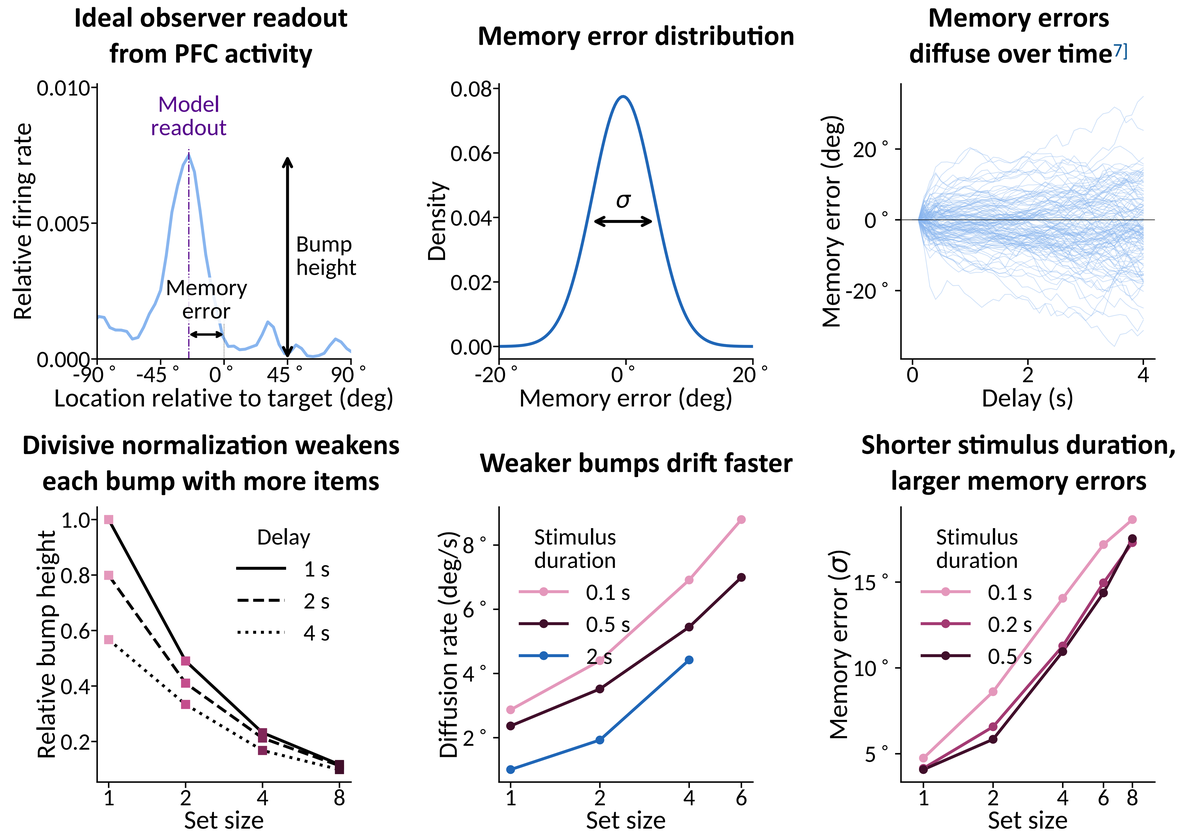
Working memory task



Model fits experimental measurements of spatial and featural WM performance



Circuit components explain visual WM dynamics



Ref. [1] Schneegans & Bays, *J. Neurosci.*, (2018); [2] Shin et al., *J. Vis.*, (2017); [3] Pertzov et al., J. *Exp. Psychol. Learn. Mem. Cogn.*, (2017); [4] Bays, *J. Neurosci.*, (2014); [5] Heeger & Mackey, *PNAS*, (2019); [6] Pal et al., *eLife*, (2026); [7] Einstein, *Ann. Phys.*, (1905).